



AI-Powered Smart Inquiry Triage Assistant



Agenda

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Why automated triage matters

1

Hundreds of inquiries arrive daily across web, app, configurator and dealer systems each read and sorted by hand before any work starts.

3

Urgency is judged case by case, so a safety-related message can sit in the queue behind routine questions.

2

Routing is inconsistent. Two agents sort the same message differently and inconsistent decisions can cause unnecessary transfers.

4

Target value: faster first-touch routing, one consistent rule, and inspectable decision evidence.

Automotive relevance: The same support channels receive routine appointment requests, payment disputes, order questions, and potentially urgent vehicle concerns. These require different teams and response priorities.



Value for Response Time and Customer Experience

Response time

- Aims to reduce manual sorting and preparation before an agent starts handling the inquiry.
- Provides a suggested queue and next actions to help the appropriate team respond sooner.

Consistency

- Applies the same taxonomy, priority-voting rules, and queue mapping across inquiries.
- Displays supporting evidence so agents can review and correct recommendations.

Customer experience

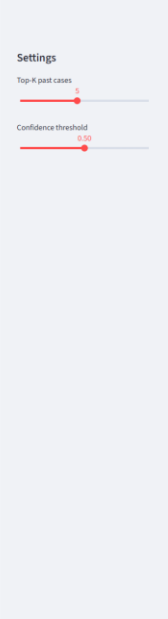
- Aims to reduce unnecessary transfers and repeated explanations.
- Helps agents focus on resolving the customer's issue rather than preparing the initial assessment.



What it produces

One inquiry in, the same six fields out every time.

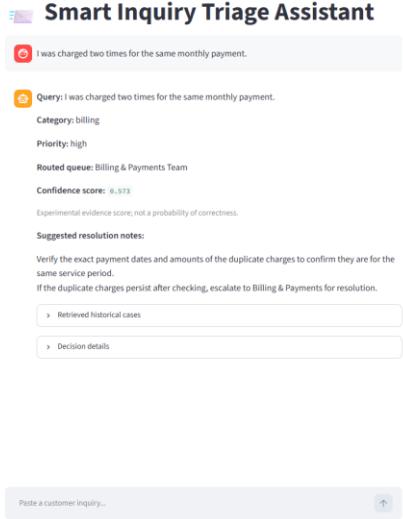
Category	one of eight taxonomy labels
Priority	low · medium · high
Routed queue	from the category mapping
Confidence	evidence score, 0 to 1
Resolution notes	one or two suggested next actions
Retrieved cases	the past inquiries used as evidence



Settings

Top-K past cases

Confidence threshold



Smart Inquiry Triage Assistant

I was charged two times for the same monthly payment.

Query: I was charged two times for the same monthly payment.

Category: billing

Priority: high

Routed queue: Billing & Payments Team

Confidence score: 0.573

Experimental evidence score; not a probability of correctness.

Suggested resolution notes:

Verify the exact payment dates and amounts of the duplicate charges to confirm they are for the same service period.

If the duplicate charges persist after checking, escalate to Billing & Payments for resolution.

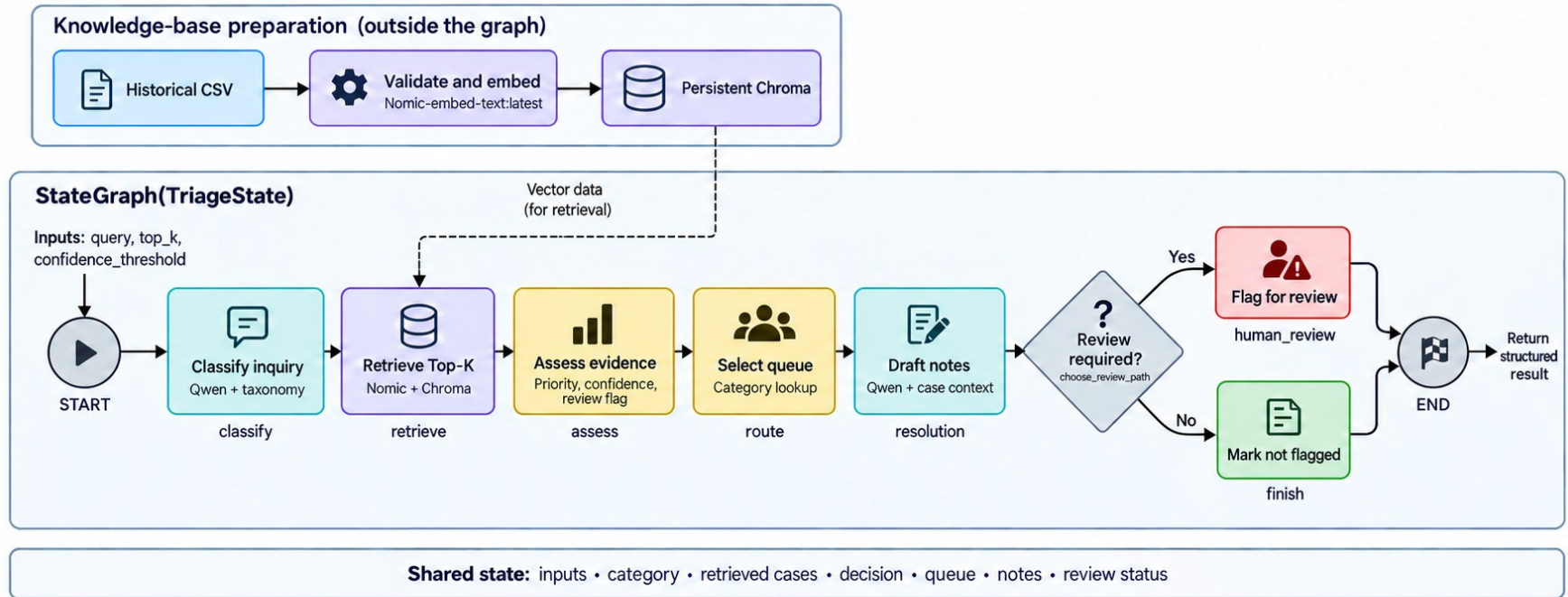
> Retrieved historical cases

> Decision details

Paste a customer inquiry...

300 historical inquiries · 8 categories · one queue each · 146 low, 100 medium, 54 high

Architecture



Review flag: confidence < threshold, or no positive matching evidence.
Final branches set status only.

Chat-model call

Embedding + retrieval

Python rules



Key decisions and trade-offs

Choice	Reason	Trade-off
Local <i>qwen3:4b</i> through Ollama	Classification and drafting without a hosted inference API	Limited instruction following and answer quality
Embedding Model <i>nomic-embed-text:latest</i> with Chroma	Finds related wording and reuses stored historical embeddings	Semantic similarity can miss differences in urgency or negation
One document per inquiry	The historical messages are short, so splitting is unnecessary	Short messages may provide limited context
Retrieval across all categories	Exposes evidence that agrees or disagrees with classification	Some retrieved neighbours may be irrelevant
Pydantic structured Output	Restricts labels and validates the output structure	Valid structure does not establish correctness



Confidence and review

A hybrid evidence score using similarity strength, category support, and priority agreement.

Why multiply: Weak relevance or conflicting labels reduce the score. The trade-off is conservative scoring and potentially frequent review.

Inquiry: “I was charged two times for the same monthly payment” at Top-K = 5

Similarity strength

0.716

Mean similarity across all
five retrieved cases

Category agreement

0.80

Four of the five retrieved
cases match billing with
positive similarity

Priority agreement

1.00

All supporting priority
weight favours high

Confidence $\approx 0.716 \times 0.80 \times 1.00 = 0.573$; Threshold 0.5: not flagged ; Threshold 0.8: flagged for review

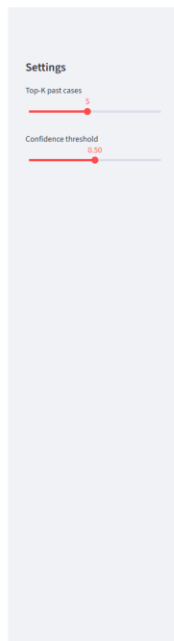


Live demonstration

What fails, and why

Three reproduced failures share one root cause: Priority errors arise from historical labels (low, medium, high); Unsuitable notes arise from insufficient safeguards.

- 01 | Negation ignored – reschedule rated high
- 02 | Real urgency under-rated – steam rated medium
- 03 | Scope not enforced – off-topic request answered
- 04 | Conservative scoring can create review workload
- 05 | Not calibrated – the score is not a probability



Smart Inquiry Triage Assistant

There is no overheating problem; I only need to change my service appointment.

Query: There is no overheating problem; I only need to change my service appointment.

Category: service

Priority: high

Routed queue: Service Scheduling Team

Confidence score: 0.387

Experimental evidence score; not a probability of correctness.

Suggested resolution notes:

Confirm the current service appointment date and time to ensure it aligns with the customer's schedule.
If the appointment is scheduled but the customer needs to reschedule, request their preferred new date and time.

Human review required. Suggested actions need verification.

Confidence is below the selected threshold.

> Retrieved historical cases

> Decision details

Paste a customer inquiry...



Prototype results and business implications

Three recorded manual checks using Top-K = 5 ; Threshold = 0.5

Exact inquiry	Recorded result	Observation
"Please send a receipt for the payment I made yesterday."	billing · medium · 0.400 · review required	Correctly identifies billing and selects Billing & Payments. Notes suggest transaction checks but do not explicitly provide the receipt.
"Something went wrong with my payment."	billing · medium · 0.279 · review required	Identifies the broad topic, but assumes a failed payment instead of clarifying what happened.
"Where should I submit an internship application?"	other · low · 0.369 · review required	Selects General Correspondence / Triage Queue. Notes suggest careers information but assume an internship portal.

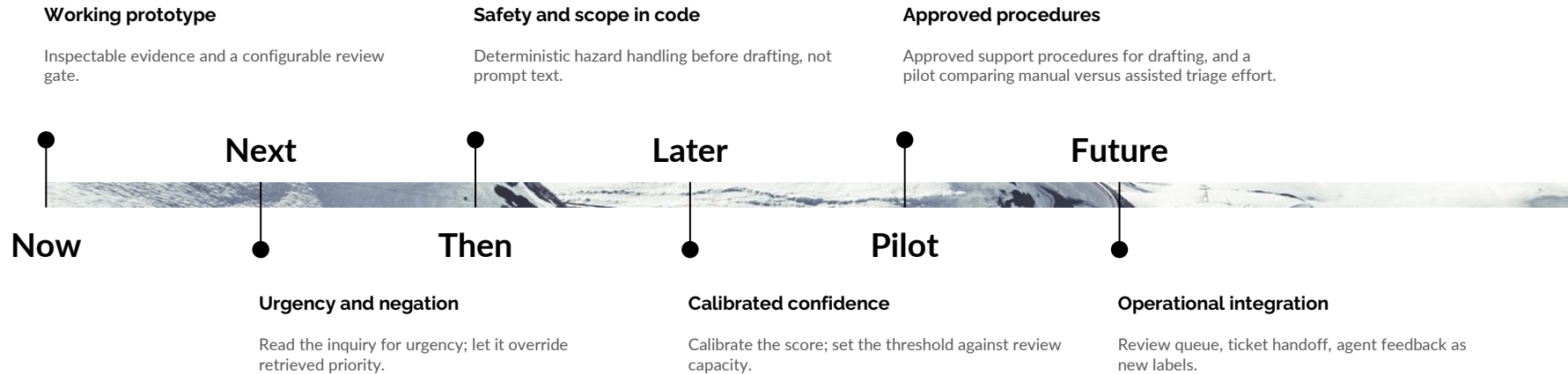
Business implication: The system prepares an assessment for an agent. It has not yet demonstrated lower total handling time or better customer outcomes.

Pilot measures: Time to correct routing, agent effort including corrections, urgent-case recall, and review workload.

Exploratory observations, not an accuracy benchmark. Generation time is not customer response time.



Roadmap





Summary

- The prototype prepares a structured triage assessment with visible historical evidence.
- Deterministic rules make priority, routing, and review decisions inspectable.
- Production adoption requires validated decision quality and demonstrated benefits for support agents.

Expected customer benefits include fewer transfers, less repeated explanation, and faster access to the appropriate team.



Thank you.

